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Iniziato	Thursday, 13 January 2022, 15:14
Stato	Completato
Terminato	Thursday, 13 January 2022, 15:31
Tempo impiegato	17 min. 14 secondi
Punteggio	15,00/15,00
Valutazione	30,00 su un massimo di 30,00 (100%)

Domanda **1**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which is the main reason for the *standardization* of numeric attributes?

~~Scegli un'alternativa:~~

-  ~~~~ a. Map all the numeric attributes to a new range such that the mean is zero and the variance is one.
- b. Remove non-standard values
- c. Change the distribution of the numeric attributes, in order to obtain gaussian distributions
- d. Map all the nominal attributes to the same range, in order to prevent the values with higher frequency from having prevailing influence

Domanda **2**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following *is not* an objective of feature selection

Scegli un'alternativa:

- ☒ a. Avoid the *curse of dimensionality*
- ☒ b. Select the features with higher range, which have more influence on the computations
- ☐ c. Reduce the effect of noise
- ☐ d. Reduce time and memory complexity of the mining algorithms

Domanda **3**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following statements is *true*?

Scegli una o più alternative:

- ☐ a. The data which are similar to the majority are never noise
- ☐ b. The noise always generate outliers
- ☒ c. The noise can generate outliers
- ☒ d. Outliers can be due to noise

Domanda **4**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In which mining activity the *Information Gain* can be useful?

Scegli un'alternativa:

- ☐ a. Clustering
- ☒ b. Classification
- ☐ c. Discretization
- ☐ d. Discovery of association rules

Your answer is correct

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

What is the *Gini Index*?

Scegli un'alternativa:

- ☐ a. A measure of the *entropy* of a dataset
- ☐ b. An accuracy measure of a dataset alternative to the *Information Gain* and to the *Misclassification Index*
- ☐ c. An impurity measure of a dataset alternative to *overfitting* and *underfitting*
- ☒ d. An impurity measure of a dataset alternative to the *Information Gain* and to the *Misclassification Index*

Domanda **6**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In a decision tree, an attribute which is used only in nodes near the leaves...

Scegli un'alternativa:

- ☒ a. ...is irrelevant with respect to the target
- ☒ b. ...gives little insight with respect to the target
- ☐ c. ...guarantees high increment of purity
- ☐ d. ...has a high correlation with respect to the target

Domanda **7**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which is the main purpose of *smoothing* in Bayesian classification?

Scegli un'alternativa:

- ☒ a. Classifying an object containing attribute values which are missing from some classes in the training set
- ☐ b. Classifying an object containing attribute values which are missing from some classes in the test set
- ☐ c. Reduce the variability of the data
- ☐ d. Dealing with missing values

Domanda 8

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

With reference to the total *sum of squared errors* and *separation* of a clustering scheme, which of the statements below is true?

- ☒ a. They are strictly correlated, if, changing the clustering scheme, one increases, then the other does the same
- ☒ b. They are strictly correlated, if, changing the clustering scheme, one increases, then the other decreases
- ☐ c. They are two ways to measure the same thing
- ☐ d. It is possible to optimise them (i.e. minimise SSE and maximise SSB) separately

Domanda 9

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

What does K-means try to minimise?

Scegli un'alternativa:

- ☐ a. The *separation*, that is the sum of the squared distances of each cluster centroid with respect to the global centroid of the dataset
- ☐ b. The *separation*, that is the sum of the squared distances of each point with respect to its centroid
- ☒ c. The *distortion*, that is the sum of the squared distances of each point with respect to the points of the other clusters
- ☒ d. The *distortion*, that is the sum of the squared distances of each point with respect to its centroid

Domanda **10**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the statements below is true? (One or more)

Scegli una o più alternative:

- ☒ ☒ Sometimes DBSCAN stops to a configuration which does not include any cluster
- ☐ b. DBSCAN always stops to a configuration which gives the optimal number of clusters
- ☒ ☒ Increasing the radius of the neighbourhood can decrease the number of noise points
- ☒ ☒ DBSCAN can give good performance when clusters have concavities

Domanda 11

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Match the rule evaluation formulas with their names

✓ $\frac{\text{conf}(A \Rightarrow C)}{\text{sup}(C)}$ LIFT

✓ $\frac{\text{sup}(A \Rightarrow C)}{\text{sup}(A)}$ CONFIDENCE

✓ $\frac{1 - \text{sup}(C)}{1 - \text{conf}(A \Rightarrow C)}$ CONVICTION

✓ $\text{sup}(A \cup C) - \text{sup}(A)\text{sup}(C)$ LEVERAGE

Domanda **12**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Vai a...

Consider the transactional dataset below

[Machine Learning - Python Lab ►](#)**ID Items**

- 1 A,B,C
- 2 A,B,D
- 3 B,D,E
- 4 C,D
- 5 A,C,D,E

Which is the *confidence* of the rule $A,C \Rightarrow B$?

$$\frac{ACB}{AC}$$

Scegli un'alternativa:

- ☒ a. 50%
- ☐ b. 40%
- ☐ c. 100%
- ☐ d. 20%

Domanda **13**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In a dataset with D attributes, how many subsets of attributes should be considered for feature selection according to an exhaustive search?

Scegli un'alternativa:

- a. $O(D)$
- b. $O(D^2)$
-   c. $O(2^D)$
- d. $O(D!)$

Domanda **14**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

How can we measure the quality of a trained regression model?

- a. With a confusion matrix
- b. Counting the number of values correctly forecast
- c. With precision, recall and accuracy
-   d. With a formula elaborating the difference between the forecast values and the true ones

Domanda **15**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which is different from the others?

Scegli un'alternativa:

- a. Apriori
-   b. Decision Tree
- c. K-means
- d. Expectation Maximisation